

TECHFACT PRO Student Course Management System



Project Description

TechFact Pro is a Python-based Student Course Management System developed as a Final Integrated Project under the Hunarmand Punjab Training Program.

The system manages students, courses, enrollments, searching, updating, deleting records and student profiles using File Handling and Object-Oriented Programming concepts.

FINAL INTEGRATED PROJECT

Version 1.0

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Hunarmand Punjab Python
Programming Course

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FINAL INTEGRATED PROJECT REPORT

PROJECT OVERVIEW

Introduction

Technology is transforming the education sector by making learning management more organized and efficient. Educational institutions require systems that can manage student records, course information, enrollments, and academic data effectively.

To address this need, a console-based application named **TechFact Pro, Student Course Management System** was developed using Python programming language. This project was created as a Final Integrated Project during the Hunarmand Punjab Python Programming Training Program.

The purpose of this project is to provide a centralized platform where student information, course records, and enrollment details can be managed efficiently through a simple menu-driven system.

The project combines various Python concepts including Object-Oriented Programming (OOP), File Handling, Functions, Modules, Loops, Conditional Statements, and Exception Handling to create a real-world management application.

PROJECT OBJECTIVES

The primary objectives of this project are:

- To develop a practical Student Management System.
 - To implement Object-Oriented Programming concepts.
 - To demonstrate file handling operations using text files.
 - To manage student and course records efficiently.
 - To perform searching, updating, and deleting operations.
 - To implement enrollment management functionality.
 - To build a real-world project using Python.
 - To improve problem-solving and programming skills.
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TECHNOLOGIES USED

The following technologies and concepts were used during the development of this project:

Programming Language

- Python 3.x

Core Concepts

- Variables and Data Types
- Operators
- Conditional Statements
- Loops
- Functions
- Lists and Strings
- Modules
- File Handling
- Exception Handling
- Object-Oriented Programming

Storage System

- Text Files (.txt)
-

PROJECT STRUCTURE

The project follows a modular structure to improve readability, maintainability, and scalability.

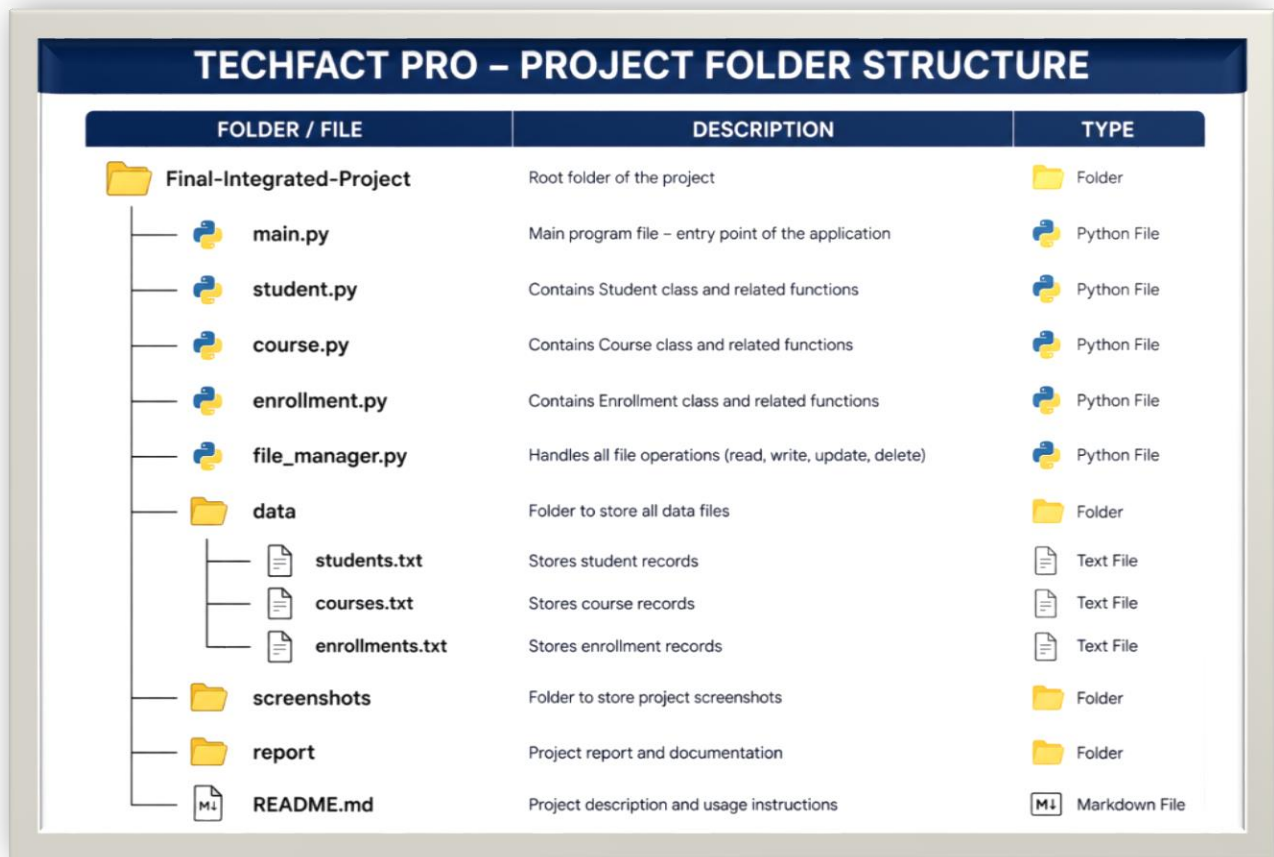


Figure 1: Project Directory Structure

TECHFACT PRO Student Course Management System

SYSTEM WORKFLOW

The system follows a simple workflow:

1. Administrator adds students.
2. Administrator creates courses.
3. Students are enrolled in courses.
4. Data is stored inside text files.
5. Users can search, update, or delete records.
6. Student profiles display complete enrollment information.

This workflow simulates a basic educational management system used in real-world institutions.

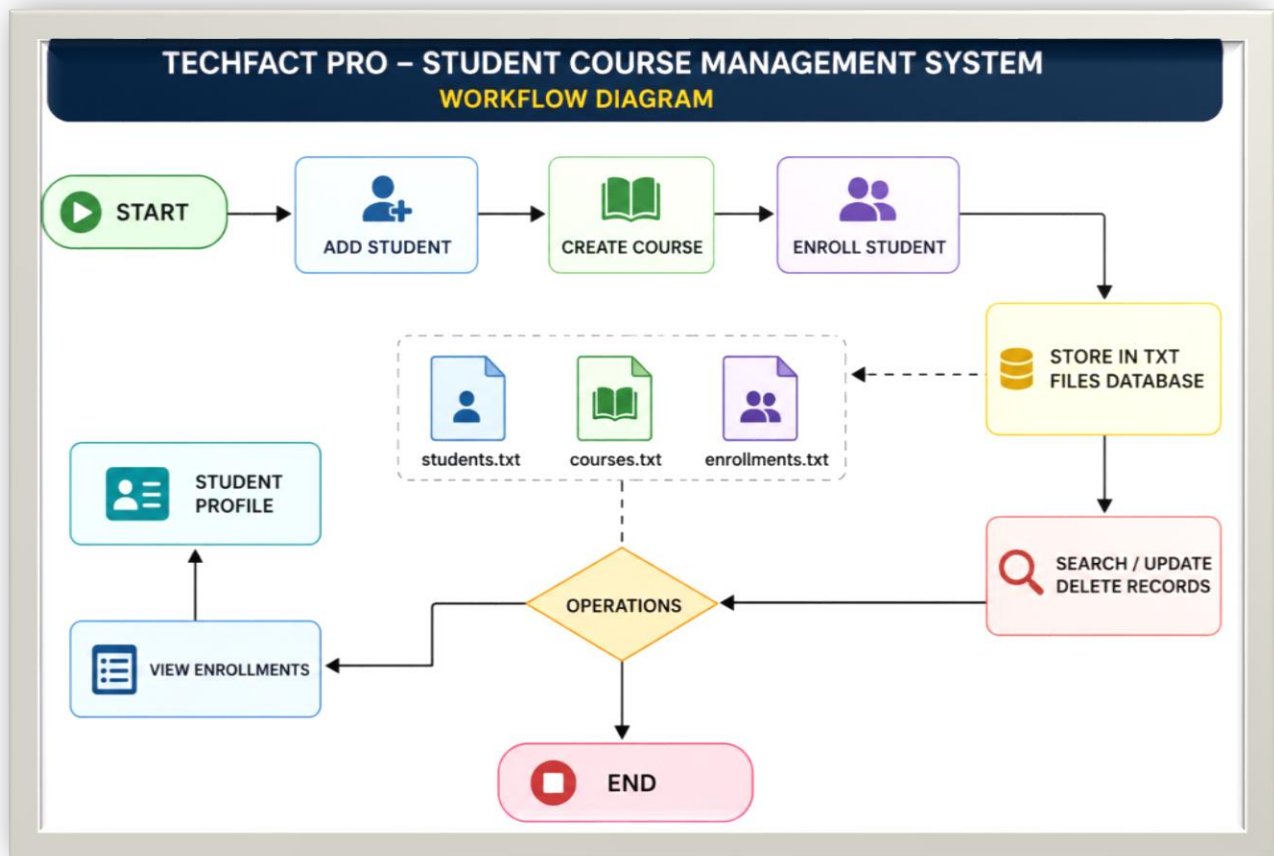


Figure 2: System Workflow Diagram

TECHFACT PRO Student Course Management System

SYSTEM ARCHITECTURE

The architecture illustrates the interaction between users, management modules, and data storage files.

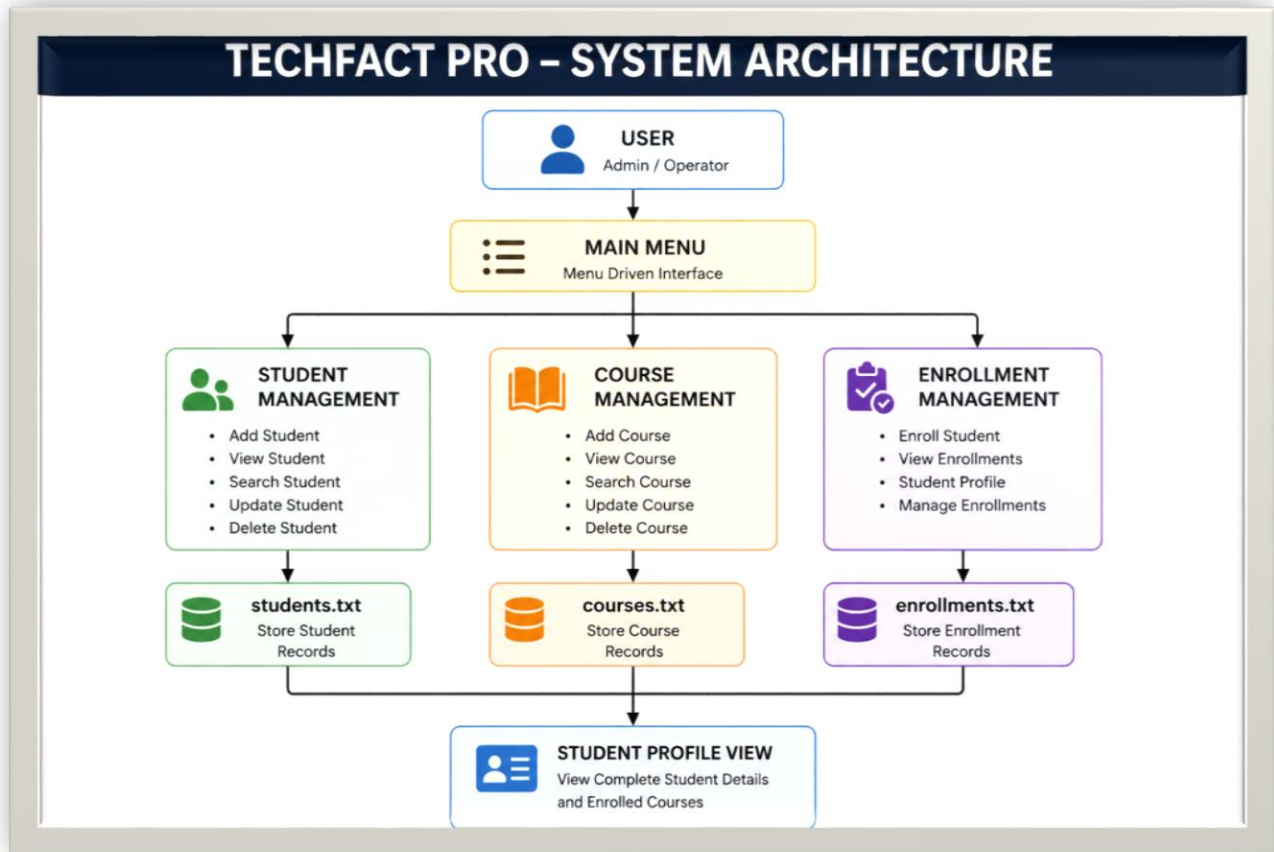


Figure 3: System Architecture Diagram

STUDENT MANAGEMENT MODULE

The Student Management Module is responsible for handling all student-related operations.

➤ **Add Student**

This feature allows users to add new student records by entering:

- Student ID
- Student Name
- Student Age

The information is stored inside the students.txt file.

➤ **View Students**

Displays all available student records saved in the system.

➤ **Search Student**

Allows users to search a student using Student ID.

➤ **Update Student**

Updates existing student information.

➤ **Delete Student**

Removes student records from the system.

Benefits

- Easy student management
 - Organized data storage
 - Fast searching and updating
-

COURSE MANAGEMENT MODULE

The Course Management Module handles all course-related operations.

➤ **Add Course**

Allows users to create new courses.

➤ **View Courses**

Displays all available courses.

➤ **Search Course**

Searches courses using Course ID.

➤ **Update Course**

Updates course information.

➤ **Delete Course**

Removes courses from the system.

Benefits

- Efficient course management
 - Organized course records
 - Easy maintenance of academic programs
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ENROLLMENT MANAGEMENT MODULE

Enrollment Management is one of the most important features of the system.

It creates a relationship between students and courses.

➤ **Enroll Student**

Students can be enrolled in one or multiple courses.

The system stores:

- Student ID
- Course ID

inside the enrollments file.

➤ **View Enrollments**

Displays all enrollment records along with student and course information.

➤ **Student Profile**

Displays complete student details including:

- Student ID
- Student Name
- Student Age
- Enrolled Courses

This feature provides a complete overview of a student's academic profile.

OBJECT-ORIENTED PROGRAMMING IMPLEMENTATION

Object-Oriented Programming was used to improve code organization and maintainability.

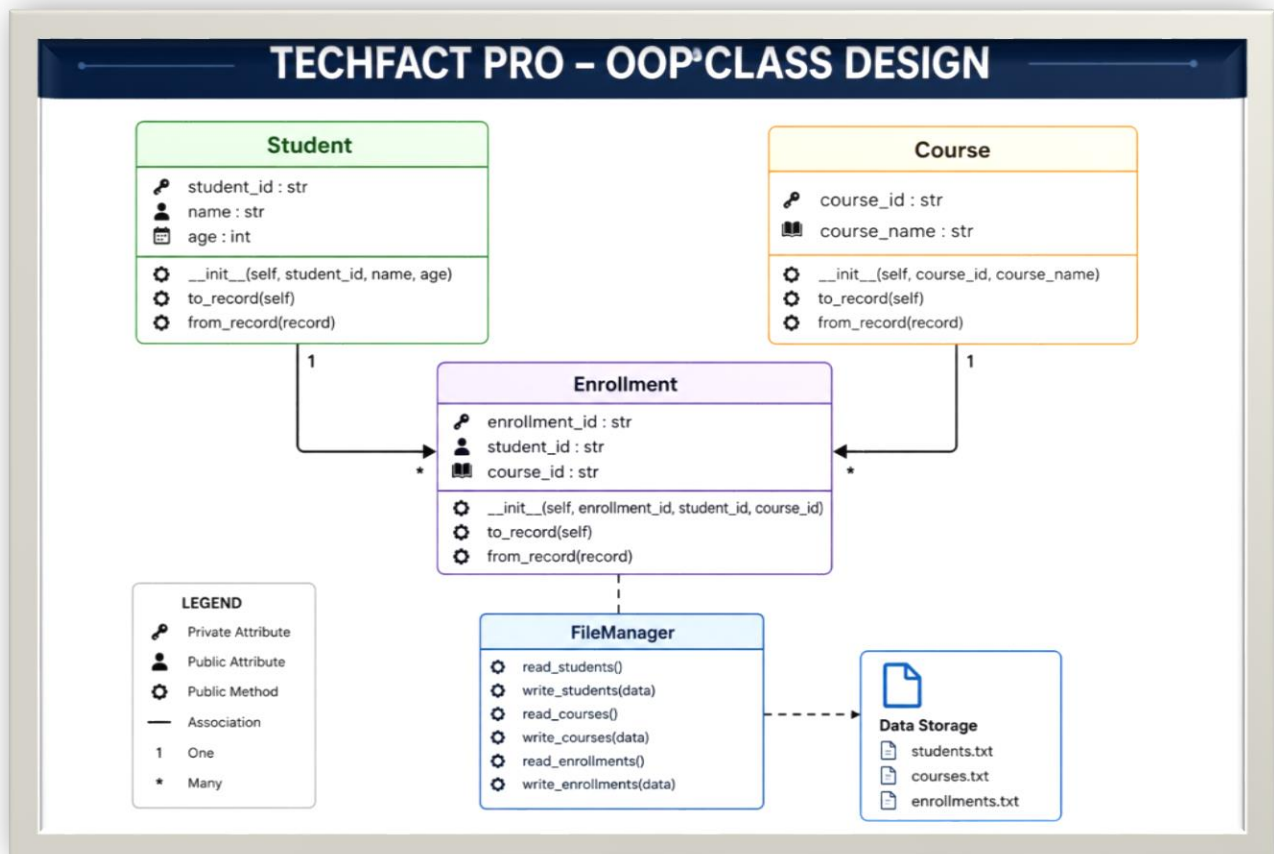


Figure 4: OOP Class Design Diagram

TECHFACT PRO Student Course Management System

➤ **Student Class**

Stores:

- Student ID
- Name
- Age

➤ **Course Class**

Stores:

- Course ID
- Course Name

➤ **Enrollment Class**

Stores:

- Student ID
- Course ID

Advantages of OOP

- Better code organization
 - Reusability
 - Maintainability
 - Scalability
 - Real-world object representation
-

TECHFACT PRO Student Course Management System

FILE HANDLING IMPLEMENTATION

The project uses text files as a simple database.

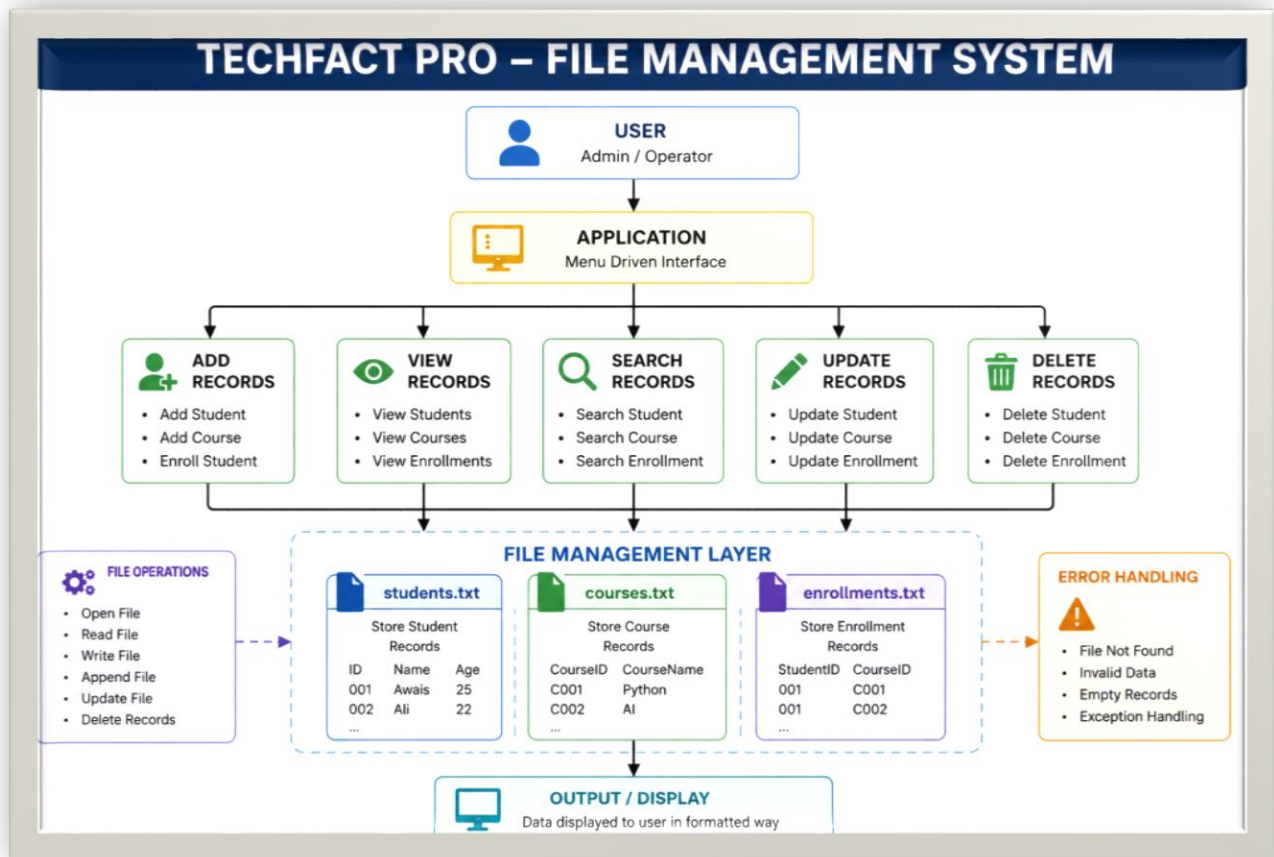


Figure 5: File Management System Architecture

TECHFACT PRO Student Course Management System

➤ **students.txt**

Stores student information.

Example:

001,Awais,25

➤ **courses.txt**

Stores course information.

Example:

C0001,Artificial Intelligence

➤ **enrollments.txt**

Stores enrollment information.

Example:

001,C0001

File Operations Used

- Open File
 - Read File
 - Write File
 - Append File
 - Update File
 - Delete Records
-

EXCEPTION HANDLING

Exception handling was implemented to avoid program crashes.

The system handles situations such as:

- Missing files
- Invalid searches
- Empty records
- Incorrect user input

This improves system reliability and user experience.

CHALLENGES FACED DURING DEVELOPMENT

Several challenges were encountered during development.

➤ **Data Relationships**

Managing relationships between students and courses required careful planning.

➤ **File Management**

Updating and deleting records from text files required custom logic.

➤ **Enrollment System**

Building the enrollment system involved linking multiple data files together.

➤ **Profile Generation**

Generating complete student profiles required data retrieval from different files.

These challenges helped improve programming and analytical skills.

LEARNING OUTCOMES

Through this project, the following skills were developed:

- Python Programming
 - Problem Solving
 - Object-Oriented Programming
 - File Handling
 - Data Management
 - Software Design
 - Modular Programming
 - Exception Handling
 - Real-World Application Development
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FUTURE IMPROVEMENTS (VERSION 2)

Future versions of the project may include:

- Dashboard Summary
- Course Analytics
- Search Student by Name
- Duplicate Record Validation
- Improved User Interface
- Report Generation
- Statistics Dashboard
- Data Export Features

These enhancements will make the system more professional and industry-ready.

GITHUB REPOSITORY

Repository:

<https://github.com/captainawais/Hunarmand-Punjab-Python-Mastery-Portfolio>

LinkedIn:

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CONCLUSION

TechFact Pro, Student Course Management System successfully demonstrates the practical implementation of Python programming concepts learned during the Hunarmand Punjab Python Programming Training Program.

The project integrates Student Management, Course Management, Enrollment Management, File Handling, Exception Handling, and Object-Oriented Programming into a complete working application.

This project not only fulfills academic requirements but also provides valuable experience in designing and developing real-world software solutions. It serves as a strong foundation for future advanced management systems and reflects the technical skills acquired throughout the training program.

DEVELOPED BY

Developed By: Awais Farooq

Program: Hunarmand Punjab – Python Programming

Instructor: Mr. Akbar Ali

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Year: 2026